THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Assertion (A): A body can have zero displacement but non-zero distance. Reason (R): Distance is the length of the path and displacement is the net change in position. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. Assertion (A): A cricketer moves his hands backward while catching a fast ball. Reason (R): This increases the time of stopping and hence decreases the force on the hands. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. Assertion (A): No work is done when a person pushes against a rigid wall. Reason (R): The displacement of the wall is zero. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. Assertion (A): Sound cannot travel in a vacuum. Reason (R): Sound waves need a material medium for propagation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A body is thrown upward with velocity u. The velocity at half the maximum height is: 1

(a) u / 2 **(b)** u **(c)** u / sqrt(2) **(d)** u / 4

6. A loaded truck of mass 4000 kg and an empty truck of mass 1000 kg are pushed by equal forces. The ratio of their accelerations (loaded : empty) is: 1

(a) 4 : 1 **(b)** 1 : 1 **(c)** 1 : 4 **(d)** 2 : 1

7. A body is thrown upward with KE of 200 J. At half the maximum height the KE is (ignoring air resistance): 1

(a) 0 **(b)** 50 J **(c)** 200 J **(d)** 100 J

8. A sound wave travels from air (340 m/s) into water (1500 m/s). A 500 Hz wave has wavelengths in air and water of: 1

(a) 0.68 m and 0.68 m **(b)** 0.68 m and 3 m
(c) 3 m and 3 m **(d)** 3 m and 0.68 m

9. A boat goes 12 km downstream in 1 hour and returns upstream in 2 hours. The average speed for the round trip is: 1

(a) 8 km/h **(b)** 6 km/h **(c)** 9 km/h **(d)** 12 km/h

10. A book is placed on a table. The reaction to the weight of the book is: 1

(a) the pull of the book on the Earth **(b)** friction
(c) the push of the table **(d)** the normal force by the table

11. A block of mass 5 kg is pulled up a smooth incline of length 10 m to a height of 5 m. The work done against gravity is (g = 10): 1

(a) 500 J **(b)** 250 J **(c)** 50 J **(d)** 25 J

12. If the temperature rises from 0 degrees Celsius to 40 degrees Celsius, the speed of sound in air increases from 331 m/s by about (0.6 m/s per degree Celsius): 1

(a) 6 m/s **(b)** 24 m/s **(c)** 40 m/s **(d)** 240 m/s

13. **Assertion (A):** The displacement of a body can be zero even though the distance travelled is not zero. 1

Reason (R): Displacement is the shortest straight-line distance from the initial position to the final position, in a given direction.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** Passengers in a bus jerk forward when the bus stops suddenly. 1

Reason (R): The lower part of the body stops with the bus, while the upper part tends to remain in motion because of inertia.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** No work is done by a person who holds a heavy suitcase while standing still. 1

Reason (R): Work is done only when a force produces a displacement of the body in the direction of the force.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A car travels the first half of the distance at 30 km/h and the next half at 60 km/h. Find the average speed. 2

17. A body of mass 2 kg moving with 10 m/s is acted on by a force of 4 N opposite to its motion. Find the time to stop and the distance. 2

18. A body of mass 2 kg is acted on by a force that changes its speed from 3 m/s to 5 m/s. Find the work done by the force. 2

19. A boy standing 85 m in front of a hill claps and hears the echo after 0.5 s. Find the speed of sound. 2

20. A body is dropped from rest and also another thrown down with 10 m/s from the same height. Compare the time of fall if h = 20 m (g = 10). 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A particle starts from rest with a = 4 m/s squared for 5 s, then moves uniformly for 10 s and then decelerates at 2 m/s squared until it stops. Find the total distance. 3

22. A car of mass 1500 kg is moving on a level road at 72 km/h. The driver applies the brakes and the car stops in 5 s. Find the force and the distance. Compare the force with the weight. 3

23. A block of mass 5 kg slides down a rough incline of height 4 m and reaches the bottom at 6 m/s. Find the work done against friction (g = 10). 3

24. Two sound waves have frequencies 100 Hz and 400 Hz with equal amplitude. Compare their wavelengths, speeds, loudness and pitch in the same medium. 3

OR

A ball rolls down an incline from rest and covers 2 m in the 1st second. Find the acceleration and the distance in 4 s. 3

25. A bullet of mass 10 g moving at 400 m/s enters a wooden block and stops in 0.04 m. Find the average retarding force and the time of penetration. 3

OR

A spring stores 4 J of energy when stretched by 2 cm. If the energy is proportional to the square of the extension, how much energy does it store when stretched by 4 cm? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A metro train in Delhi starts from rest at a station and accelerates uniformly at 0.5 m/s^2 for 40 s. It then moves at constant velocity for the next 60 s. 5

(i) Find the velocity of the train after 40 s.
(ii) Find the distance covered in the first 40 s.
(iii) Find the distance covered in the next 60 s.
(iv) Find the total distance covered in the 100 s and the average speed over this time.

27. A block of mass 2 kg is released from rest at the top of a smooth curved track of height 3.2 m, slides along a rough horizontal surface of length 4 m and stops. Find the speed at the bottom, the friction force and the work done by friction (g = 10). 5

OR

A source of frequency 440 Hz sends a sound wave in air (344 m/s) towards a wall 172 m away. Find the wavelength, the time for the echo, the number of waves between the source and the wall, and the number of complete waves in the echo path. 5

END OF QUESTION PAPER — Total: 50 Marks